

Mixed Review: Logarithms → Polynomials — Solutions

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Part 1 — Logarithms

Problem 1

Use change-of-base to evaluate $\log_5(28)$. Exact + 4 decimals.

Change of Base

Log Evaluation

Step 1. Apply change-of-base formula: $\log_b(a) = \frac{\ln a}{\ln b}$.

Step 2. Exact: $\log_5(28) = \frac{\ln 28}{\ln 5} = \frac{\log 28}{\log 5}$.

Step 3. Decimal: $\frac{3.33220}{1.60944} \approx 2.07039$.

$$\log_5(28) = \frac{\ln 28}{\ln 5} \approx 2.0704$$

Problem 2

Expand $\log_3\left(\frac{x^2\sqrt{y}}{z^4}\right)$.

Quotient Rule

Product Rule

Power Rule

Step 1. Quotient rule: $\log_3(x^2\sqrt{y}) - \log_3(z^4)$.

Step 2. Product rule on numerator: $\log_3(x^2) + \log_3(\sqrt{y}) - \log_3(z^4)$.

Step 3. Rewrite radical: $\sqrt{y} = y^{1/2}$. Power rule on each term:

$$2\log_3(x) + \frac{1}{2}\log_3(y) - 4\log_3(z)$$

$$2\log_3(x) + \frac{1}{2}\log_3(y) - 4\log_3(z)$$

Problem 3

Condense $3 \ln(x) - \frac{1}{2} \ln(y+1) + 2 \ln(z)$.

Power Rule (reverse)

Combine into Single Log

Step 1. Reverse power rule: $\ln(x^3) - \ln((y+1)^{1/2}) + \ln(z^2)$.

Step 2. Combine sum: $\ln(x^3 z^2) - \ln(\sqrt{y+1})$.

Step 3. Combine difference into quotient.

$$\ln\left(\frac{x^3 z^2}{\sqrt{y+1}}\right)$$

Problem 4

Solve $\log_2(x-3) + \log_2(x+1) = 5$. State domain; reject extraneous.

Log Equation

Domain Restriction

Extraneous Check

Step 1 (Domain). Need $x-3 > 0$ AND $x+1 > 0 \Rightarrow x > 3$.

Step 2. Combine logs: $\log_2((x-3)(x+1)) = 5$.

Step 3. Convert to exponential form: $(x-3)(x+1) = 2^5 = 32$.

Step 4. Expand: $x^2 - 2x - 3 = 32 \Rightarrow x^2 - 2x - 35 = 0$.

Step 5. Factor: $(x-7)(x+5) = 0 \Rightarrow x = 7$ or $x = -5$.

Step 6 (Reject). $x = -5$ violates $x > 3$ (extraneous). Keep $x = 7$.

$$x = 7$$

Part 2 — Exponential Modeling

Problem 5

\$4,500 at 4.2% continuously compounded. When reaches \$10,000?

Continuous Compound Interest

$$A = Pe^{rt}$$

Step 1. Set up: $10,000 = 4,500 e^{0.042 t}$.

Step 2. Isolate exponential: $\frac{10,000}{4,500} = \frac{20}{9} = e^{0.042 t}$.

Step 3. Take natural log: $\ln\left(\frac{20}{9}\right) = 0.042 t$.

Step 4. Solve: $t = \frac{\ln(20/9)}{0.042} = \frac{0.79851}{0.042} \approx 19.012$.

$$t \approx 19.01 \text{ years}$$

Problem 6

$A(t) = A_0 e^{kt}$, half-life 28 yr. (a) Find k . (b) Time until 12% remains.

Half-Life

Exponential Decay

ln Manipulation

Part (a) Step 1. At $t = 28$, $A = \frac{1}{2} A_0$: $\frac{1}{2} A_0 = A_0 e^{28k}$.

Step 2. Cancel A_0 : $\frac{1}{2} = e^{28k}$.

Step 3. Take ln: $\ln\left(\frac{1}{2}\right) = 28k \Rightarrow k = -\frac{\ln 2}{28}$.

Decimal: $k \approx -0.02476$ per year.

Part (b) Step 1. Set $0.12 A_0 = A_0 e^{kt} \Rightarrow 0.12 = e^{kt}$.

Step 2. Take ln: $\ln(0.12) = kt$.

Step 3. Substitute k : $t = \frac{\ln(0.12)}{-\ln(2)/28} = -\frac{28 \ln(0.12)}{\ln(2)}$.

Step 4. Decimal: $t = -\frac{28(-2.1203)}{0.6931} \approx \frac{59.37}{0.6931} \approx 85.66$.

$$\text{(a) } k = -\frac{\ln 2}{28} \approx -0.0248 \quad \text{(b) } t \approx 85.66 \text{ years}$$

Problem 7

Rewrite $f(t) = 600(1.035)^t$ as $600e^{rt}$. Find r and interpret.

Base Conversion

ln Identity

Step 1. Equate exponentials: $(1.035)^t = e^{rt}$.

Step 2. Use identity $a^t = e^{t \ln a}$: $e^{t \ln(1.035)} = e^{rt}$.

Step 3. Match exponents: $r = \ln(1.035)$.

Step 4. Decimal: $r \approx 0.03440$, so the continuous growth rate is $\approx 3.44\%$ per unit of t — slightly less than the discrete 3.5% , since continuous growth compounds infinitely often.

$$r = \ln(1.035) \approx 0.0344 \Rightarrow \approx 3.44\% \text{ continuous}$$

Part 3 — Fundamental Theorem of Algebra

Problem 8

$(2 - i)$ is a zero of $P(x) = x^4 - 5x^3 + 7x^2 + 3x - 10$. Find all zeros and write linear factored form.

Conjugate Root Theorem

Polynomial Long Division

Step 1. Real coefficients $\Rightarrow$ complex zeros come in conjugate pairs. So $(2 + i)$ is also a zero.

Step 2. Build the quadratic factor:

$$(x - (2 - i))(x - (2 + i)) = x^2 - 4x + (4 - i^2) = x^2 - 4x + 5.$$

Step 3. Divide $P(x)$ by $x^2 - 4x + 5$:

$$\frac{x^4 - 5x^3 + 7x^2 + 3x - 10}{x^2 - 4x + 5} = x^2 - x - 2.$$

Step 4. Factor remaining quadratic: $x^2 - x - 2 = (x - 2)(x + 1)$.

Step 5. All zeros: $x = 2, -1, 2 - i, 2 + i$.

$$P(x) = (x - 2)(x + 1)(x - (2 - i))(x - (2 + i))$$

Problem 9

Degree 4, real integer coefficients. Zeros: $x = 3$ (mult. 2), $x = 1 + 2i$. Standard form.

Conjugate Root Theorem

Polynomial Construction

Step 1. Real coefs $\Rightarrow 1 - 2i$ is also a zero. Four zeros: 3, 3, $1 + 2i$, $1 - 2i$.

Step 2. Build:

$$P(x) = (x - 3)^2(x - (1 + 2i))(x - (1 - 2i)).$$

Step 3. Multiply complex pair:

$$(x - (1 + 2i))(x - (1 - 2i)) = (x - 1)^2 - (2i)^2 = (x - 1)^2 + 4 = x^2 - 2x + 5.$$

Step 4. Multiply $(x - 3)^2 = x^2 - 6x + 9$ with $x^2 - 2x + 5$:

$$(x^2 - 6x + 9)(x^2 - 2x + 5) = x^4 - 8x^3 + 26x^2 - 48x + 45.$$

$$P(x) = x^4 - 8x^3 + 26x^2 - 48x + 45$$

Part 4 — Synthetic Division & Factor Theorem

Problem 10

Synthetic division: $(3x^4 - 8x^3 + 5x - 7) \div (x - 2)$. Verify with Remainder Theorem.

Synthetic Division

Remainder Theorem

Step 1. Coefficients (insert 0 for missing x^2): 3, -8, 0, 5, -7. Divisor root: $c = 2$.

Step 2. Synthetic division table:

2	3	-8	0	5	-7
		6	-4	-8	-6
	3	-2	-4	-3	-13

Step 3. Read off: quotient $Q(x) = 3x^3 - 2x^2 - 4x - 3$, remainder $R = -13$.

Step 4 (Verify). By Remainder Theorem, $P(2)$ should equal R :

$$P(2) = 3(16) - 8(8) + 0 + 5(2) - 7 = 48 - 64 + 10 - 7 = -13. \checkmark$$

$$Q(x) = 3x^3 - 2x^2 - 4x - 3, \quad R = -13$$

Problem 11

Find k so $(x + 4)$ is a factor of $P(x) = 2x^3 + kx^2 - 11x - 12$.

Factor Theorem

Solve for Parameter

Step 1 (Factor Theorem). $(x + 4)$ is a factor $\iff P(-4) = 0$.

Step 2. Substitute $x = -4$:

$$P(-4) = 2(-4)^3 + k(-4)^2 - 11(-4) - 12 = -128 + 16k + 44 - 12.$$

Step 3. Simplify: $P(-4) = 16k - 96$.

Step 4. Set to zero: $16k - 96 = 0 \Rightarrow k = 6$.

$$k = 6$$

Part 5 — Polynomial Graphing

Problem 12

$P(x) = -(x+1)^2(x-2)^3(x-4)$. Degree, end behavior, zeros & multiplicity, y -intercept, sketch.

Multiplicity

Bounce / Cross / Slide

End Behavior

(a) Degree & End Behavior: Sum of multiplicities $= 2 + 3 + 1 = 6$. Leading term comes from $-x^2 \cdot x^3 \cdot x = -x^6$. Degree even & leading coefficient negative $\Rightarrow$ both ends point down.

As $x \rightarrow \pm\infty$, $P(x) \rightarrow -\infty$. $\downarrow\downarrow$

(b) Zeros & Multiplicity:

- $x = -1$, multiplicity 2 (even) $\rightarrow$ **bounce**
- $x = 2$, multiplicity 3 (odd, > 1) $\rightarrow$ **slide / flat-cross**
- $x = 4$, multiplicity 1 $\rightarrow$ **cross** (linear)

(c) y -intercept: $P(0) = -(1)^2(-2)^3(-4) = -(1)(-8)(-4) = -32$. Point: $(0, -32)$.

(d) Sketch summary (sign chart):

Interval	$(-\infty, -1)$	$(-1, 2)$	$(2, 4)$	$(4, \infty)$
Sign of P	—	—	+	—

Curve: comes up from $-\infty$ on the left $\rightarrow$ *touches and bounces* at $x = -1$ (stays below) $\rightarrow$ passes through $(0, -32)$ $\rightarrow$ *flat-crosses* through $x = 2$ (rises above axis) $\rightarrow$ peaks $\rightarrow$ *crosses* at $x = 4$ (drops back below) $\rightarrow -\infty$ on the right.

Degree 6, ends $\downarrow\downarrow$; zeros: -1 (bounce), 2 (slide), 4 (cross); y -int $(0, -32)$